Feiyang Liu

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Research Interests

Language-Centric Multimodal Systems: Unified Multimodal Pretraining; Scalable and Controllable Multimodal Models; Solving Vision Problems with LLMs; Symbolic Vision; End-to-End Multimodal Training Systems.

Education

South China University of Technology Sep 2022 – Jun 2027 (expected)
B.Eng. in Intelligent Science and Technology

University of California, Berkeley Aug 2024 – May 2025
Visiting Student

Publications

1. Haoquan Zhang, **Feiyang Liu**, Juanxi Tian, Wanhua Li, Haodong Duan, Weiyang Liu. **PixelBench: Can LLMs Understand and Generate Images Pixel by Pixel?** 2026. [Project]
2. Qinzheng Zhou, **Feiyang Liu**, Lijing Lu, Zhihang Li. **Confidence-Guided Multi-Scale Aggregation for Sparse-View High-Resolution 3D Gaussian Splatting.** *CVPR 2026*, **Highlight**. [Paper]
3. Qinzheng Zhou, **Feiyang Liu**, Lijing Lu, Zhihang Li. **Commit Only When Justified: Counterfactual Topology Selection for Sparse-View 3DGS.** Under review (Phase 2), *AAAI 2027*.

Research Experience

OpenUMM: Demystifying the Design Space of Unified Multimodal Pretraining Ongoing
Project Lead; UC Santa Cruz and Gemini Team

- Leading a controlled study of **unified multimodal pretraining** (in the spirit of *Towards Physics of Multimodal Pretraining* and *Beyond Language Modeling*) unifying language modeling, visual understanding, and image generation (Qwen3 + flow matching), ablating **visual tokenizers/encoders**, **architecture/MoT routing**, **pretraining data mixtures/resolution**, and **cross-task synergy**.
- Investigating **multi-stage pretraining recipes** (warm-up, module learning rates, text replay, loss weighting) and **performance prediction**, correlating pretraining signals and short probes with post-training and final outcomes.
- Built and optimized a modular JAX/Flax pretraining, data, and decontaminated evaluation stack across **hundreds of TPU v5p/v6e chips** (multi-host/multi-slice sharding, ZeRO-1, rematerialization, elastic recovery, XLA profiling).

GroundingBench Ongoing
Research Contributor; CUHK and ByteDance Seed

- Investigating whether modern LLMs can directly replace specialist and generative vision models on standard computer vision tasks, in contrast to approaches that repurpose image or video generation models for visual perception (e.g., Vision Banana, GenCeption, and SenseNova-Vision).
- Responsible for a subset of 2D visual tasks (including object detection and counting): curating and verifying training and evaluation data, defining evaluation protocols, and tuning and evaluating baseline models.

PixelBench
Research Contributor; CUHK and ByteDance Seed

- Probed LLMs as general visual generators without agentic tools, visual tokenizers, or diffusion modules, generating pixel images, 3D voxels, and video in text as **symbolic visual primitives** and scaling with coding/reasoning ability.
- Co-developed **Pixel Reconstruction Tuning (PRT)** in pretraining/post-training using text-encoded pixels to boost visual understanding; analyzed visual abstraction (downsampling, resolution) and built the workbench.

3D Reconstruction
Co-author; HUST

- Contributed algorithm design and writing for confidence-guided multi-scale reconstruction (CVPR Highlight); co-authored follow-up research on counterfactual topology selection for sparse-view 3DGS (AAAI 2027 Phase 2).

Technical Skills

- **ML Systems:** JAX/Flax, PyTorch, multi-host/multi-slice TPU training, sharding, ZeRO-1, XLA profiling.
- **Graphics & 3D:** C++ graphics programming, path tracing, mesh processing, physical simulation, 3DGS; Three.js / React Three Fiber, Blender, OpenGL/GLSL, Unity.
- **Programming:** Python, C/C++, Java.